SULIT

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION
SEMESTER I 2022/2023

CODE OF SUBJECT	**: SECJ3323**
NAME OF SUBJECT	**: SOFTWARE ARCHITECTURE AND DESIGN**
YEAR / COURSE	**: 1/2/3/4SECJ**
TIME	**: 3 Hours**
DATE	**:22nd February 2023/(WEDNESDAY)**
VENUE	**:**

THIS EXAMINATION BOOK CONSISTS OF TWO (2) PARTS:

- **PART A**: 5 STRUCTURED QUESTIONS (50 MARKS)
  Answer all the questions
- **PART B**: 2 PROBLEM SOLVING QUESTION (50 MARKS)
  Answer all the questions

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET PROVIDED.

Name	
I/C number	
Year/Course	
Section	
Lecturer	

This Examination book consist of 9 printed pages excluding this page

PART A: STRUCTURED QUESTION **[50 Marks]**

Question 1 **[18 marks]**

a. PDL (Programming Design Language) popularity stems from the use of natural languages as opposed to computer languages or graphical techniques to define the required behaviour of functions. Based on the Shift Schedule module below, write up a PDL that show which result in detailed function designs that are easier to create, easier to review, and easier to translate to code.

[8 marks]

Staff Scheduling Access Level: The system needs to differentiate the different level of hotel positions. Only the staff admin of the hotel can create and update the staff schedule and assign every task to each staff. Schedule Management: The staff admin need to consider a few things to create a balance and good staff schedule such as considering the occupancy levels and events expected in the hotel, the spread of duty hours, whether 12 hours, 16 hours or 24 hours, decide whether the positions will work for 5 or 7 days per week, type of shift such as shift-straight shift, break shift, rotating shift, or any other alternative scheduling-to be used. Other requirements that need to be considered are the number of full and part time staff on the payroll. Figure 1 shows the flowchart of the workflow to create staff schedule:

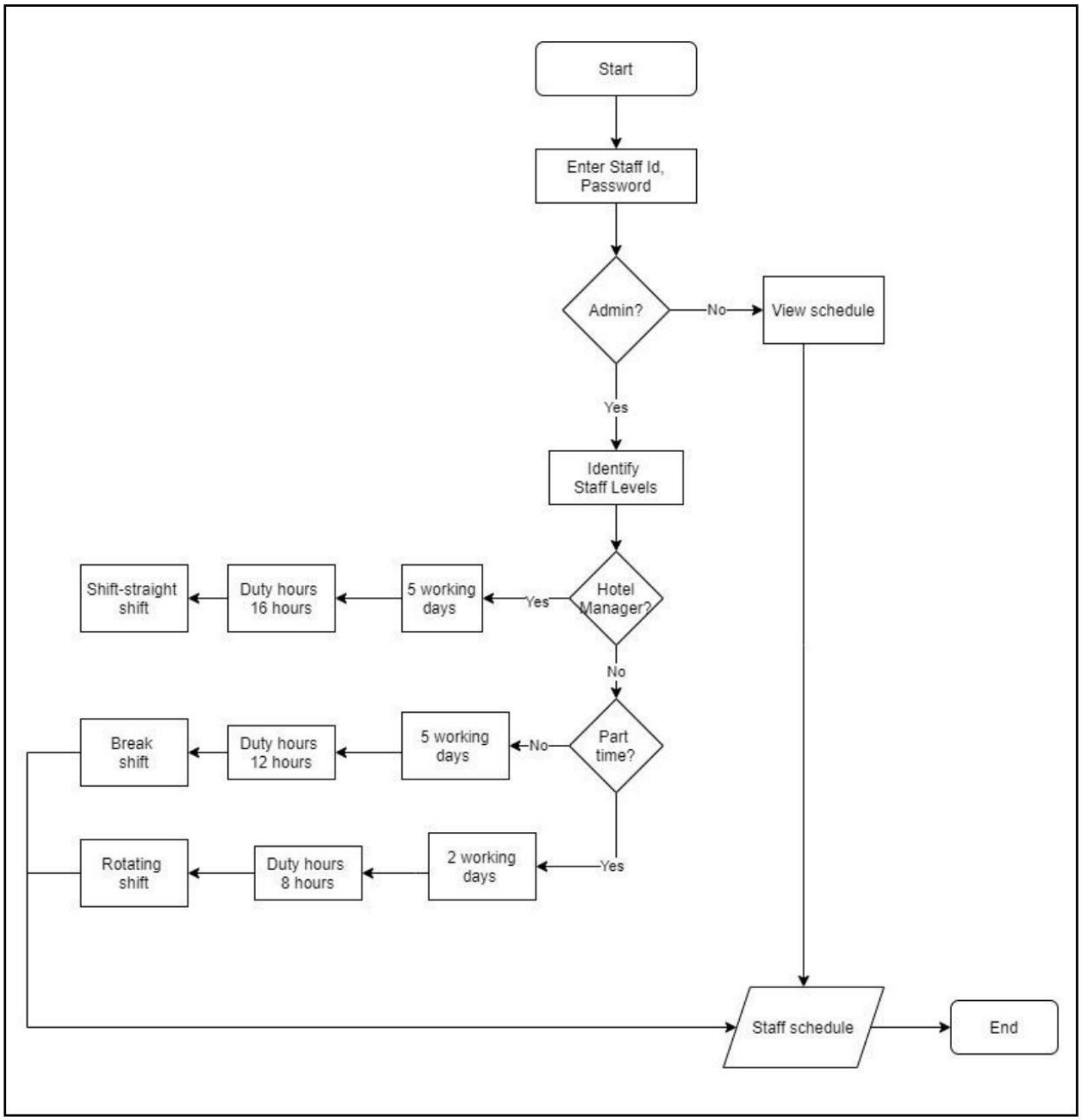


Figure 1

b. The cyclomatic complexity computation allows designers to measure the complexity of flow-based operational designs by determining the complexity of the decision structure of operations instead of lines of code. Based on Figure 2, convert the codes to draw the program flow Graph and determine the cyclomatic complexity using ONE (1) of the approaches that you have learned.

[10 marks]

```cpp
#include <iostream>

using namespace std;

int main()
{
 char repeat;
 int passenger;
 int age;
double tickectprice;
 do
 {
     cout<<"Please insert the number of passenger(s)"<<endl;
     cin>>passenger;
     tickectprice=0.00;
     for (int a=1;a<=passenger;a++)
     {
        mistake: cout<<"Please insert the age for passenger no "<<a<<endl;
         cin>>age;
         if (age<0 || age>100 )
         goto mistake;
         if (age<15)
         {cout<<"Underage is not allowed for this ride. You need to re-enter again the no. of passenger"<<endl;
         break;}
         tickectprice+=25.00;

     }
     cout<<"Would you like to add passengers? if yes please press y"<<endl;
     cin>>repeat;
 } while (repeat=='y'||repeat=='Y');
 cout<<"The total price for the ride is RM "<<tickectprice;

    system("pause");
    return 0;
}
```

Figure 2

Question 2 **[10 marks]**

MediNET Usage Invoicing: The supplier (ASP) produces invoices on MediNET usage to both HCPs and paymasters. HCPs are charged based on the following two-tier fee structure in Table 1:

Table 1: Two-tier fee structure

Monthly Subscription Fee	**RM 50.00** per month
Transaction Fee	**RM 0.15** per transaction

For example, let assume that a General Practice that uses the MediNET application to treat a total of 400 patients for the month of February 2004. Thus, the fee would be calculated as follows:

Table 2: Fee Calculation

No	Description	Calculation	Fee Payable
1	Monthly Subscription Fee	RM 50.00 x 1 month	RM 50.00
2	Transaction Fee	RM 0.15 x 400 transactions	RM60.00
		Total fee for the month	**RM 110.00**

For the paymasters, they will be charged based on the number of their payees. The following table shows the fee structure for the paymasters:

Table 3:Medinet System Document Fee

Employee + Dependant Amount Range (person) *		Monthly Subsciption Fee (RM)
Min	**Max**	
1	25	**35**
26	50	**65**
51	75	**95**
76	100	**125**
101	150	**165**
151	200	**200**
201	300	**250**
301	400	**300**
401	500	**375**
501	600	**450**
601	700	**525**
701	800	**600**
801	900	**675**
901	1000	**725**
1000	~	**negotiable**

Based on the MediNET System Description document fees structure in Figure 2, draw an EEDT (Extended Entry Decision Table). Give an excerpt of the code for the implementation of the drawn EEDT.

[10 marks]

Question 3 **[12 marks]**

Answer Question 3 based on the Descriptions given below:

1. There are 5 states: requested, confirmed, on waiting list, used and cancelled for the status of the book room booking.
2. The room is confirmed if the room is available, the room count will be decrease.
3. The customer will be put on waiting list if the room is not available.
4. The room state is changed to used after the customer moves in.
5. The room state is changed to archived after the customer check out and pays. The room count increased.
6. The state is cancelled if the customer cancel the room or the customer give up the room on waiting list. The room count increased.

Figure 3

Draw a state diagram for the above scenario in Figure 3. **[12 marks]**

Question 4 **[10 marks]**

a. In the design phase, management refers to activities required to efficiently create quality design artifacts, within schedule and budget constraints

i. Based on the following information, calculate the Earned Value (EV), Cost Performance Index (CPI), Schedule Performance Index (SPI), Estimated Remaining Cost, and Estimated Total Cost (ETC). Please refer to Table 1 in appendix A as guideline in answering this question. **[10 marks]**

- Number of task is a design project: 5 tasks
- Estimated cost of each task: RM 2,000
- Total budget for the project (BAC): RM 10,000
- Estimated duration to complete each task: 1 month
- Estimated total planned duration for the project: 5 months
- In the second month (i.e. current month), only one task have been fully completed at a cost of RM3,000

PART B: PROBLEM SOLVING [50 Marks]

Part B consists of **TWO (2)** essay questions. Answer **all the** questions. Each question carries 25 marks.

Question 1 [30 marks]

i. Apply suitable design pattern to the following design problem in Figure 5 and give **TWO (2)** justifications on why the design pattern is selected and describe **TWO (2)** benefits of using that design pattern. **[18 marks]**

The XYZ company has developed a Spool Tracking System (STS), enable the workers at the ship yard to track a spool using RFID tag and mobile apps that being processed for production. STS can do the following function:

initiate()
trackSpool()
registerSpoolID()
scanSpool()
end()

As an appointed consultant, you have to improve the STS class so that the *trackSpool(), registerSpoolID* and *scanSpool()* may perform using different tag (QR Code). However, it is a time taking task to modify the code and luckily, you manage to find the code with the third party company for QR code mechanism. The third party code consists of the following operations; *trackSpoolQR(), registerSpoolIDQR()* and *scanSpoolQR()*. Thus you want to reuse the class diagram and code to improve the existing STS. STS can use the functions from the third party code (QR).

Figure 5

i. Based on the Ayu Villa Hotel System descriptions below, apply suitable design pattern and draw a class diagram based on chosen pattern. **[12 marks]**

Ayu Villa Hotel System has a Customer Services module subsystem that handle the Promotion and Vouchers that hotel offers: the system can provide promotional notification that allows travel agencies, customers and staff to receive notifications for vouchers and discounts (1 notification per minute) via SMS and email.The system is based on anomaly notification engine that interface with several vouchers and promotions data sources (gift vouchers, breakfast vouchers, room promotions advertisment) where these data will be fused and analysed and automatically decide what kind of package, types of rooms and other facilities that customer frequently choose to provide promotions and vouchers. Hotel Ayu Villa is interested to develop a decision engine, which is responsible to send a notification for discounts and vouchers to the customers via existing communication channel (SMS and email). The design must be flexible so that other type of communication channels which can be added in the near future

Question 2 **[20 marks]**

i. The following questions describe the additional requirements that require you to apply the suitable design patterns. Each of the following questions does not relate to each other. The diagram in Figure 6 shows the information flow for generating different representation for Purchase Order form for the Procurement Subsystem. Which design pattern is suitable Figure 6 and draw the class diagram that applies your selected pattern, and describe **TWO (2)** justifications to support your design pattern selection.

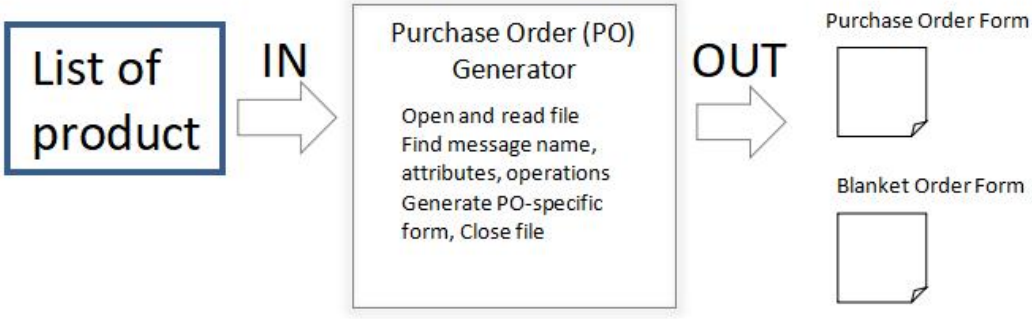


Figure 6

[10 marks]

ii. Write the implementation code using any programming language (C++, Java, Phyton, etc) based on the class diagram drawn in Question 2ai.

[10 marks]

- END QUESTION -

APPENDIX A

Table 1:Earned Value Management Calculation Table

Parameter	Equation
EV	EV = (% work completed) * planned cost for the work
CV	CV = EV - AC
SV	SV = EV - PV
CPI	CPI = EV / AC
SPI	SPI = EV / PV
ETC	ETC = (BAC – EV) / CPI
EAC	EAC = ETC + AC